



-
1. In RSA, if user A wants to send an encrypted message to user B, the plaintext is encrypted with the public key of ____
- A. User A
B. User B
C. The network
D. A third party.
-
2. In a ____ the relationship between a character in the plaintext to a character in the ciphertext is one-to-many
- A. Monoalphabetic cipher
B. Substitution cipher
C. Polyalphabetic cipher
D. Transposition cipher
-
3. In the object-oriented paradigm, as in nature ____ is an object that can inherit from another object.
- A. Method
B. Inheritance
C. Polymorphism
D. Classes
-
4. The first generation of computers available was based on the ____ bits micro processors.
- A. 4
B. 8
C. 16
D. 64
E. None of the others
-
5. _____
- B. 145.12



The octal equivalent of 1100101.001010 is ____

- A. 624.12
- B. 145.12
- C. 154.12
- D. 145.21
- E. None of the others

6. A group includes ____ bits in the binary system are represented as one digit in the hexadecimal system. D. 4

- A. 1
- B. 2
- C. 3
- D. 4
- E. None of the others

7. What is the field of Natural Language Processing (NLP)? D. All of the above

- A. Computer Science
- B. Artificial Intelligence
- C. Linguistics
- D. All of the above
- E. None of the above

8. The devices in Bluetooth LAN, sometimes called gadgets, find each other and make a network called ____ D. A piconet

- A. ESS
- B. Ad hoc
- C. BSS
- D. A piconet

9. What is the field that investigates the mechanics of human intelligence? B. cognitive science

- A. history



- B. cognitive science
- C. psychology
- D. sociology

-
10. ___ is an English-language-like representation of an algorithm. There is no standard for it- some people use a lot of detail
- A. Pseudocode
- B. Unified Modeling Language (UML)
- C. Flowchart
- D. State diagram
-
11. In ___ the development process flows in only one direction. This means that a phase cannot be started until the previous phase is completed.
- A. Waterfall model
- B. Incremental model
- C. Development proces
- D. Software engineering
-
12. According to the von Neumann model, ___ are stored in memory.
- A. Only data
- B. Only programs
- C. Data and programs
- D. Neither data nor programs
-
13. In physical layer, an ___ has infinitely many levels of intensity over a period of time
- A. Analog signa
- IsB. Electromagnetic signals
- C. Digital signals
- D. Electronic signals



-
14. The ___ of each node includes one of its children and all descendents of that child B. Binary tree
- A. Left tree
 - B. Binary tree
 - C. Subtree
 - D. Right tree
-
15. In the sign-and-magnitude representation, the memory location can hold four bits, assume that we have stored integer 6 in a memory location and we then try to add 6 to the integer. The computer's response is ___? D. -4
- A. 12
 - B. -12
 - C. 4
 - D. -4
-
16. The ___ level of a three-level DBMS architecture defines the logical view of the data. B. Conceptual
- A. External
 - B. Conceptual
 - C. Internal
 - D. Physical
-
17. The ___ is a protocol that is used to define how the client-server programs can be written to retrieve web pages from the Web. D. HTTP
- A. FTP
 - B. SMTP
 - C. TELNET
 - D. HTTP
-



-
18. Artificial Intelligence has evolved extremely in all the fields except for ____ E. None of the above
- A. Web mining
 - B. Construction of plans in real time dynamic systems
 - C. Understanding natural language robustly
 - D. All of the above
 - E. None of the above
-
19. A program that is employed in the development, repair or enhancement of other programs is known as B. Software tool
- A. System software
 - B. Software tool
 - C. Applications program
 - D. Utility program
 - E. None of the others
-
20. Which of following represents a set of data items that share a specific relationship? B. Data structure
- A. ADT
 - B. Data structure
 - C. Programming language
 - D. Data element
-
21. In a positional number system with base b , we can always find the number of digits of an integer. So how many digits can we find in the decimal number 234 in decimal? C. 3
- A. 1
 - B. 2
 - C. 3
 - D. 4
-
- 22.



___ means that the sender and the receiver expect confidentiality.

D. encryption and decryption

A. Nonrepudiation

B. Integrity

C. Authentication

D. encryption and decryption

23. An electronic circuit with about 20 transistors fabricated on a silicon chip is known as

A. SSI

B. MSI

C. DPS

D. RJE

E. None of the others

24. The ___ acts as a liaison between a client program and a server program, a process-to-process connection.

D. Transport

A. Physical

B. Data-link

C. Network

D. Transport

25. Which is a unit representing the no bits of discrete.

A

A. Baud

B. Byte

C. Bit

D. All of the others

E. None of the others

26. ___ is a program that facilitates the execution of other programs.

A

A. An operating system

B. Hardware



- C. A queue
- D. An application program

-
27. Each column in a relation is called ____ A
- A. An attribute
 - B. A tuple
 - C. A union
 - D. An attitude
-
28. The device manager will be tracking every device constantly to ensure that the device is ____ C. Functioning properly
- A. Ready
 - B. Finished
 - C. Functioning properly
 - D. Running
-
29. ____ is a method in which each statement in the software is executed at least once. A
- A. Basis path testing
 - B. Condition testing
 - C. Loop testing
 - D. Boundary testing
-
30. Which of the following representations is erroneous? D
- A. (110)₂
 - B. (141)₈
 - C. (EF)₁₆
 - D. 22_A
-
31. If the memory address space is 16 MB and the word size is 8 bits, then ____ bits are needed to access each word. C
- A. 8



- B. 16
- C. 24
- D. 32

-
32. The push operation ___ of the stack. C
- A. Deletes an item from the top
 - B. Deletes an item from the bottom
 - C. Inserts an item at the top
 - D. Inserts an item at the bottom
-
33. In the ___ graphic method of representing an image in a computer, re-scaling of the image creates a ragged or grainy image. A
- A. Bitmap
 - B. Vector
 - C. Quantized
 - D. Binary
-
34. In the network database model, the entities are organized in a ___ in which some entities can be accessed through several paths A
- A. Graph
 - B. Tree
 - C. Array
 - D. Record
-
35. In ___ method to solve collision the first record is stored in the home address but contains a pointer to the second record. A
- A. Linked list resolution
 - B. Open addressing resolution
 - C. Bucket hashing
 - D. Digit extraction hashing



-
36. In ___ edges are ordered pairs of vertices. If x and y are two vertices and an edge $e = (x, y)$, x is called the starting point and y is called the endpoint of the edge. A
- A. Directed graph
 - B. Undirected graph
 - C. Edges
 - D. Arcs
-
37. A hashed file uses a ___ to accomplish this mapping D
- A. Retrived file
 - B. Sequential access
 - C. Random access
 - D. Mathematical function
-
38. The unit of a computer system that executes program, communicates with and often controls the operation of other subsystems of the computer is known as ___ A
- A. CPU
 - B. Control Unit
 - C. I/O unit
 - D. Peripheral unit
 - E. None of the others
-
39. Use a circular right shift operation on the bit pattern 1001 1000 A
- A. 0100 1100
 - B. 0011 0001
 - C. 1001 1001
 - D. 1001 1000
-
40. FORTRAN is a(n) ___ language. A
- A. Procedural



- B. Functional
- C. Declarative
- D. Object-oriented

-
41. Arithmetic right shift is used to __ an integer by two C
- A. Addition
 - B. Subtraction
 - C. Divide
 - D. Multiply
-
42. __ is a basic algorithm in which we want to find the location of a target in a list of items. B
- A. Sorting
 - B. Searching
 - C. Product
 - D. Summation
-
43. In the system development process, writing the program is part of the __ phase. C
- A. Analysis
 - B. Design
 - C. Implementation
 - D. Testing
-
44. The number of bits per sample is sometimes referred to as the __ D
- A. Quantization
 - B. Sampling rate
 - C. Encoding
 - D. Bit depth
-
45. All the members of a record must be __ B
- A. The same type



- B. Related types
- C. Integer type
- D. Character type

-
46. A unary operator is applied to ___ relation(s) and creates an output of ___ relation(s). A
- A. One, one
 - B. One, two
 - C. Two, one
 - D. Two, two
-
47. When converting a decimal fraction to base b, we repeatedly ___ b. B. Multiply by
- A. Divide by
 - B. Multiply by
 - C. Add to
 - D. Subtract from
-
48. In ___ only one program can reside in memory for execution. A
- A. Mono-programming
 - B. Multi-programming
 - C. Partitioning
 - D. Paging
-
49. The attack that reduces the capability of a computer is called a ___ attack. B
- A. Penetration
 - B. Denial of service
 - C. Either a or b
 - D. Neither a nor b

50.

B



When there is memory space available to load the program totally or partially the job moves to the ____.
It now becomes a process.

- A. Hold state
- B. Ready state
- C. Running state
- D. Waiting state

-
51. Which printer is very commonly used for desk-top publishing? A
- A. Laser printer
 - B. Ink-jet printer
 - C. Daisy wheel printer
 - D. Dot-matrix printer
 - E. None of the others

-
52. In ____ sort, the smallest item moves to the beginning of the unsorted list. There is no one-to-one swapping. B. Bubble
- A. Selection
 - B. Bubble
 - C. Insertion
 - D. Every

-
53. Which physical topology uses a hub or switch? C
- A. Bus
 - B. Ring
 - C. Star
 - D. Bus and ring

-
54. A ____ is a predetermined value used in a program and it may change value with time. D
- A. Constants
 - B. Literals



C. Input/output

D. Variables

55. In the __ hashing method, selected digits are extracted from the key and used as the address. D

A. Direct

B. Division remainder

C. Modulo division

D. Digit extraction

56. __ is a basic algorithm that adds a list of numbers. A

A. Summation

B. Product

C. Smallest

D. Largest

57. What's happen if the size of memory is the less than the size of the program in monoprogramming OS? D

A. Need to use virtual memory

B. No overflow

C. Overflow

D. The program cannot be run

58. The __ manager is responsible for archiving and back-up. D

A. Memory

B. Process

C. Device

D. File

59. The format of an IPv4 address is referred to as __ C

A. Dotted-binary notation

B. Dotted-octal notation



- C. Dotted-decimal notation
- D. Dotted-hexadecimal notation

-
60. A/an ___ is a program in a procedural language that is a set of statements. The statements are normally executed one after another. However, sometimes it is necessary to change this sequential order. This figure illustrates for this statement. D
- A. Assignment statement
 - B. None of others
 - C. Compound statement
 - D. Control statement
-
61. A path by which communication is achieved between a central processor and other devices is called ____ C
- A. Audit trail
 - B. Network
 - C. Bus
 - D. Channel
 - E. None of the others
-
62. ___ refers to a process that rounds the value of a sample to the closest integer value. A
- A. Quantization
 - B. Sampled
 - C. Decoded
 - D. Coded
-
63. Research scientists all over the world are taking steps towards building computers with circuits patterned after the complex interconnections existing among the human brain's nerve cells. What name is given to such type of computers? D



- A. Intelligent computers
- B. Supercomputers
- C. Smart computers
- D. Neural network computers
- E. None of the above

64. A (n) ___ is a suitable structure when a small number of insertions and deletions are required, but a lot of searching and retrieval is needed C. Linked list

- A. Array
- B. Record
- C. Linked list
- D. Tree

65. The device manager maintains one or more ___ for similar input/output devices D

- A. Graph
- B. Array
- C. Stack
- D. Queue

66. ___ is a set of characters that is treated like an array in some languages and as a type in others. A. A string

- A. A string
- B. A record
- C. A symbol
- D. A link

67. The items are divided into two lists (sorted and unsorted) ___ sort. D

- A. Only in a selection
- B. Only in a bubble



- C. Only in an insertion
- D. In selection, bubble, or insertion

-
68. A control bus with eight wires can define ___ operations. C
- A. 8
 - B. 16
 - C. 256
 - D. 512
-
69. ___ separated the programming task from computer operation tasks. C
- A. Algorithms
 - B. Data processors
 - C. High-level programming languages
 - D. Operating systems
-
70. In ___ cryptography, everyone has access to everyone's public key. B
- A. Symmetric-key
 - B. Asymmetric-key
 - C. Secret-key
 - D. Private-key
-
71. A computer assisted method for the recording and analyzing of existing or hypothetical system is B
- A. Data transmission
 - B. Data flow
 - C. Data capture
 - D. Data processing
 - E. None of the others



What is the method in black-box testing that a subset of values in the input domain is selected for testing?

- A. Boundary-value testing
- B. Random testing
- C. Exhaustive testing
- D. Data flow testing

73. Convert the binary number $(110.11)_2$ to decimal C

- A. 6.7
- B. 7.75
- C. 6.75
- D. 27

74. ___ is multi-programming with swapping. C

- A. Partitioning
- B. Paging
- C. Demand paging
- D. Queuing

75. The scanning rate in image processing is called ___. If C the resolution is sufficiently high, the human eye cannot recognize the discontinuity in reproduced images

- A. Bit map
- B. Pixels
- C. Resolution
- D. Color depth
- E. None of others

76. The Stack operation returns a/an ___ B

- A. Full stack
- B. Empty stack
- C. Check status of stack
- D. All of the others



-
77. The unary ___ operation always results in a relation that has exactly one more row than the original relation. A
- A. Insert
 - B. Delete
 - C. Update
 - D. Select
-
78. A ___ is a collection of related logically coherent data used by the application programs in an organization. A. Database
- A. Database
 - B. File
 - C. Directory
 - D. Flat-file
-
79. ___ can provide authentication, integrity, and nonrepudiation for a message. B
- A. Encryption/decryption
 - B. Digital signature
 - C. Compression
 - D. Key-exchange
-
80. The ___ scheduler moves a process from one process state to another. B
- A. Job
 - B. Process
 - C. Virtual
 - D. Queue
-
81. If A is the first data element input into a queue, followed by B, C, and D, then ___ is the first data be removed. A



- A. A
- B. B
- C. C
- D. D

-
82. The file structure allows random access is A
- A. Hashed file
 - B. Sequential file
 - C. EOF
 - D. None of others
-
83. The maximum value of a hexadecimal integer with K C digits is $N_{\max} = \underline{\hspace{1cm}}$?
- A. 16 bits
 - B. 16^K
 - C. $16^K - 1$
 - D. $2^K \cdot 2^4$
 - E. None of the others
-
84. In the a program is considered a mathematical B function. In this context a function is a maps a list of inputs to a list of outputs
- A. Procedural paradigm
 - B. Functional paradigm
 - C. Declarative paradigm
 - D. Object-oriented paradigm
-
85. RISC (reduced instruction set computer). The strategy D behind RISC architecture is to have a small set of instructions that do a minimum number of
- A. Complex instructions
 - B. Complex operations



-
- C. Simple operations
 - D. Simple instructions
-

86. Use a circular left shift operation on the bit pattern 1001 1000 B
- A. 0001 1001
 - B. 0011 0001
 - C. 1001 1001
 - D. 1001 1000
-
87. Each row in a relation is called ____ B. A tuple
- A. An attribute
 - B. A tuple
 - C. A union
 - D. An attitude
-
88. ____ is a protocol for accessing and transferring documents on the WWW. D
- A. FTP
 - B. SMTP
 - C. TELNET
 - D. HTTP
-
89. What is quality referring to the ability to move software from one hardware platform to another D. Portability
- A. Flexibility
 - B. Interoperability
 - C. Resusability
 - D. Portability
-
90. What is method in the object-oriented paradigm means that we can define several operation with the same name that can do different things in related C



classes?

- A. Method
- B. Inheritance
- C. Polymorphism
- D. Classes

91. The ___ layer of the TCP/IP protocol suite is responsible for node-to-node delivery of a frame between two adjacent nodes. C. Data-link

- A. Transport
- B. Network
- C. Data-link
- D. Session

92. Which of the following is equivalent to 12 in decimal? B

- A. (1110)₂
- B. (C)₁₆
- C. (15)₈
- D. None of the above

93. The ethical principle that dictates that an act is ethical when a majority of people in society agrees with it is referred to as C. Social contract

- A. Moral rules
- B. Utilization
- C. Social contract
- D. None of the others

94. A ___ gives the user's view of a system: it shows how the user communicates with the system C

- A. Class diagram
- B. State diagram



- C. Use-case diagram
- D. Sequence diagram

-
95. If a graph is directed, the relations between edges and arcs are ____
- A. One-way
 - B. Two-way
 - C. No-way
 - D. Any - way
-
96. ____ is a pictorial representation of an algorithm. A
- A. A UML diagram
 - B. A program
 - C. Pseudocode
 - D. An algorithm
-
97. What is the result of one's complement B? A
- A. Not B
 - B. Add B with 1
 - C. XOR B with 0
 - D. AND B with 0
 - E. OR B with 1
-
98. Before a disk drive can access any sector record, a computer program has to provide the record's disk address. What information does this address specify? D
- A. Track number
 - B. Sector number
 - C. Surface number
 - D. All of the others
 - E. None of the others

-
99. C



The physic-layer addresses, sometimes called ___ are locally defined addresses, each of which defines a specific host or router in a network (LAN or WAN).

- A. TCP
- B. Port numbers
- C. MAC addresses
- D. IP address

100. The maximum value of an octal integer with K=2 digits A is $N_{\max} = \underline{\hspace{1cm}}$

- A. 63
- B. 64
- C. 65
- D. 66
- E. None of the others

101. ___ file can be accessed randomly.

D. Any

- A. A sequential
- B. An indexed
- C. A hashed
- D. Any

102. In ___ more than one program is in memory at the same time and they are executed concurrently with the CPU switching rapidly between the programs

B

- A. Monoprogramming
- B. Multiprogramming
- C. Microprogramming
- D. Operating system

103. Of the various database models, the ___ model is the most prevalent today.

C

- A. Hierarchical



- B. Network
- C. Relational
- D. Linked list

-
104. Use a simple right shift operation on the bit pattern 1001 1000. D
- A. 1001 1001
 - B. 1001 1000
 - C. 0100 1101
 - D. 0100 1100
-
105. C++ and C# language are a(n) ___ language D
- A. Procedural
 - B. Functional
 - C. Declarative
 - D. Object-oriented
-
106. ___ is multi-programming without swapping. A
- A. Partitioning
 - B. Virtual memory
 - C. Demand paging
 - D. Queuing
-
107. The ___ in the TCP/IP protocol suite is responsible for the host-to-host delivery of messages C
- A. Physical
 - B. Data-link
 - C. Network
 - D. Transport
-
108. An indexed file consists of ___ D
- A. Only a sequential data file
 - B. Only an index



- C. Only a random data file
- D. An index and random data file

109. Pascal is a(n) ___ language.

A

- A. Procedural
- B. Functional
- C. Declarative
- D. Object-oriented

110. A(n) ___ is a sequenced collection of elements of the same data type and use index to indicate the ordinal number of the element counting from the beginning

A

- A. Array
- B. Record
- C. Linked list
- D. Tree

111. A group contains ___ bits in the binary system are represented as one digit in the octal system.

D

- A. 1
- B. 2
- C. 3
- D. 4
- E. None of the others

112. A ___ is a collection of data stored in the internal format of the computer. This contains data that is meaningful only if it is properly interpreted by a program.

B. Binary file

- A. Text file
- B. Binary file
- C. Sequential file
- D. EOF



-
113. Software quality can be divided into three broad measures C
- A. Accuracy - usability- testability
 - B. Security - operability - reusability
 - C. Reusability - maintainability- security
 - D. Maintainability - transferability- operability
-
114. A computer is a box full of electronic C
- A. Switching devices
 - B. Chips
 - C. Circuits
 - D. Registers
 - E. None of the others
-
115. A storage system for small amounts of data is C
- A. Magnetic card
 - B. Magnetic tape
 - C. Punched card
 - D. Optical mark reader
 - E. None of the others
-
116. Convert the FRACTION decimal number 0.625 to binary B
- A. 1.010
 - B. 0.101
 - C. 1.110
 - D. 0.110
-
117. In storing audio, how many samples per second is good enough to reproduce an audio signal? D
- A. 10 000
 - B. 20 000



-
- C. 30 000
 - D. 40 000
-

118. ___ is required to allow any relation in the database to be represented, to allow a query language

- A. Relational database
 - B. Database system
 - C. Atomic operations
 - D. Normalization
-

119. Which component is not an operating system component?

- A. File access management program
 - B. Device driver program
 - C. Program scheduler for process
 - D. Free memory management program
-

120. The ___ of ethics says an act is ethical if a majority of people in society agree with it.

- A. Moral rules
 - B. Social contract
 - C. Utilization
 - D. Nonmaleficence
-

121. In a binary tree, each node has ___ two subtrees. C. At most

- A. More than
 - B. Less than
 - C. At most
 - D. At least
-

122. A transport-layer payload may be ___ in several network-layer packets. A

- A. Encapsulated



- B. Decapsulated
- C. Encrypted
- D. Decrypted

-
123. ___ is the amateur. Usually their techniques are deployed out of ill motives such as revenge attacks D. Blue Hat Hacker
- A. Black Hat Hacker
 - B. Grey Hat Hacker
 - C. Red Hat Hacker
 - D. Blue Hat Hacker
-
124. ___ is the event that occurs when a hashing algorithm produces an address for an insertion key but that address is already occupied D
- A. Open addressing
 - B. Every possible key
 - C. Digit extraction key
 - D. Collision
-
125. Control Unit of a digital computer is often called the ___ C. ICs
- A. Clock
 - B. Nerve center
 - C. ICs
 - D. All of the others
 - E. None of the others
-
126. A memory that does not change its contents without external causes is known as B. Static memory
- A. Dynamic memory
 - B. Static memory
 - C. RAM



D. EEPROM

E. None of the others

127. An index register that is automatically incremented or decremented with each use is

A. Auto index

B. Asynchronous

C. Assembler

D. Compiler

E. None of the others

128. A ___ is a collection of related elements possibly of different types having a single name

A. Array

B. Record

C. Linked list

D. Tree

129. The third generation of computers covers the period:

A. 1971-1982

B. 1982-1994

C. 1959-1964

D. 1965-1971

E. 1995-now

130. In ___ sort, the item that goes into the sorted list is always the first item in the unsorted list.

A. Selection

B. Bubble

C. Insertion

D. Every

131. D



What kind of state does a process execute until it needs I/O resources when I/O is complete?

- A. Hold state
- B. Ready state
- C. Running state
- D. Waiting state

132. We can convert a number in any system to decimal. We multiply each digit with its ___ in the source system and add the result to get the number in the decimal system. C

- A. Numbers
- B. Values
- C. Place value
- D. None of others

133. ___ is a common language in the business - environment. D

- A. FORTRAN
- B. C++
- C. C
- D. COBOL

134. A number which is stored and processed but not in the standard exponential form is called B

- A. Floating point number
- B. Fixed point number
- C. Octal number
- D. Decimal number
- E. None of the others

135. Assume a new Excess system uses 17 bits to represent the exponent section. What is the bias value in this C



system?

- A. 17
- B. 16
- C. 65535
- D. 65536

136. Which of the following is not the commonly used programming language for Artificial Intelligence? A

- A. Perl
- B. Java
- C. PROLOG
- D. Python
- E. None of the above

137. In ___ cryptography, the same key is used by the sender and the receiver. A

- A. Symmetric-key
- B. Asymmetric-key
- C. Public-key
- D. Open-key

138. Two integers A and B are stored in two's complement format. Show how B is subtracted from A? B

- A. A add one's complement B
- B. A add two's complement B
- C. A subtract one's complement B
- D. A subtract two's complement B
- E. None of others

139. An expert system differs from a database program in that only an expert system ___ B

- A. contains declarative knowledge
- B. contains procedural knowledge



- C. features the retrieval of stored information
- D. expects users to draw their own conclusions
- E. None of the above

140. We use a ___ search for an unordered list. A. Sequential

- A. Sequential
- B. Binary
- C. Bubble
- D. Insertion

141. A stack is a ___ structure. B

- A. FIFO
- B. LIFO
- C. DIFO
- D. SIFO

142. ___ is a step-by-step method for solving a problem or doing a task. D

- A. A construct
- B. A recursion
- C. An iteration
- D. An algorithm

143. Computer can not do anything without a ___ D

- A. Chip
- B. Memory
- C. Output device
- D. Program
- E. None of the others

144. The ___ layer of the TCP/IP protocol suite provides services for end users. C

- A. Data-link



- B. Transport
- C. Application
- D. Physical

-
145. In CISC (complex instruction set compute), The type of programming that uses microoperations is called ____ D
- A. Microoperations
 - B. Micromemory
 - C. Microcomputer
 - D. Microprogramming
-
146. Each node in a graph can have __ parents. C
- A. Only one
 - B. No node
 - C. One or more
 - D. At most two node
-
147. The no of logical records in a physical record is B
- A. Group
 - B. Blocking factor
 - C. Sector
 - D. Field
 - E. None of the others
-
148. Symmetric-key cryptography is based on sharing se- D. Personal secrecy
crecy; asymmetric-key cryptography is based on ____
- A. Secrets
 - B. Mathematical function
 - C. Privacy data
 - D. Personal secrecy
-
149. In the software quality, what factor measures can be B. Timeliness
mentioned for operability?



- A. Flexibility
- B. Timeliness
- C. Resusability
- D. Changeability

150. In the ___ graphic method of representing an image in a computer, the image is decomposed into a combination of geometrical figures. B

- A. Bitmap
- B. Vector
- C. Quantized
- D. Binary

151. Which of the following representations is erroneous? A

- A. $(10211)_2$
- B. $(342)_8$
- C. $(EEE)_{16}$
- D. 145

152. The base of the octal number system is B

- A. 2
- B. 8
- C. 10
- D. 16

153. In Windows Operating System, folders are organized hierarchically, the top level is called root-directory. C

What is this organization called?

- A. Peer-to-peer organization
- B. Organization up and down
- C. Tree Organization
- D. Spider network Organization



-
154. Every process is ____ D
- A. Only a job
 - B. Only a program
 - C. Only a partition
 - D. A job and a program
-
155. Each element in a record is called ____ C
- A. A variable
 - B. An index
 - C. A field
 - D. A node
-
156. A control unit with five wires can define up to ____ D
- operations.
- A. 5
 - B. 10
 - C. 16
 - D. 32
-
157. Backing storage is so named because it C
- A. Is always kept at the back of the CPU.
 - B. Is slow and backward
 - C. Backs up the computer's main memory
 - D. Lags behind the main memory
 - E. None of the others
-
158. In the third generation of computers: E
- A. Distributed data processing first became popular
 - B. An operating system was first developed
 - C. High-level procedural languages were first used
 - D. Online, realtime systems first became popular
 - E. All of the others
-



-
159. ___ is an ordered set of unambiguous steps that produces a result and terminates in a finite time. D. An algorithm
- A. A construct
 - B. A recursion
 - C. An iteration
 - D. An algorithm
-
160. The ___ method of integer representation is the most common method for storing integers in computer memory. C
- A. sign-and-magnitude
 - B. one's complement
 - C. two's complement
 - D. unsigned integers
-
161. An ___ is a language-specific token that requires an action to be taken such as arithmetic B
- A. Expression
 - B. Operator
 - C. Variables
 - D. Input/output
-
162. A ___ does not substitute one symbol for another; instead it changes the location of the symbol D
- A. Monoalphabetic cipher
 - B. Substitution cipher
 - C. Polyalphabetic cipher
 - D. Transposition cipher
-
163. In the data link layer, a Bluetooth LAN is a (an) ___ B
- A. Wired WANs
 - B. Ad hoc



- C. Dial-up service
- D. Extended Service Set

-
164. The first widely-used commercial form of Artificial Intelligence (AI) is being used in many popular products like microwave ovens, automobiles and plug in circuit boards for desktop PCs. It allows machine to handle vague information with a deftness that mimics human intuition. What is the name of this Artificial Intelligence? C
- A. Boolean logic
 - B. Human logic
 - C. Fuzzy logic
 - D. Functional logic
-
165. Inside the ADT are two different parts of the model that are __ and ____ A
- A. Data structure and operations
 - B. Application program and public operations
 - C. Application program and private operations
 - D. Data struture and application program
-
166. ____ supervises the activity of each component in a computer system. A
- A. An operating system
 - B. Hardware
 - C. A queue
 - D. An application program
-
167. In postorder traversal of a binary tree, the root is processed ____ C
- A. First
 - B. Second



- C. Last
- D. After the left subtree

168. Human beings are referred to as Homosapiens. Which device is called Silico sapiens? D

- A. Monitor
- B. Hardware
- C. Robot
- D. Computer
- E. None of the others

169. Which of the following is the branch of Artificial Intelligence? A

- A. Machine Learning
- B. Cyber forensics
- C. Full-Stack Developer
- D. Network Design

170. In the ___ we can think of a program as an active agent that manipulates passive objects. D

- A. Procedural paradigm
- B. Functional paradigm
- C. Declarative paradigm
- D. Object-oriented paradigm

171. The ___ of ethics says that we should avoid doing anything if it is against universal moral standards A. Moral rules

- A. Moral rules
- B. Social contract
- C. Utilization
- D. Nonmaleficence

172. B



What is model a process of software development where requirements divided into multiple modules of the software development cycle?

- A. Waterfall model
- B. Incremental model
- C. Development process
- D. Software engineering

173. The ___ area is the file area that contains all the home addresses. D. Prime

- A. Probe
- B. Linked
- C. Hash
- D. Prime

174. The ___ model is the basis for today's computers. B

- A. Leibnitz
- B. Von Neumann
- C. Pascal
- D. Charles Babbage

175. The ___ construct handles repeated actions. C

- A. Sequence
- B. Decision
- C. Repetition
- D. Flow

176. In the storing images, the samples are called ___ (which stands for picture elements). B

- A. Bit map
- B. Pixels
- C. Resolution



- D. Color depth
- E. None of others

-
177. In direct hashing the key is the data file address ____ C
algorithmic manipulation. The file must therefore contain a record for every possible key
- A. Modulo division
 - B. Division remainder
 - C. Without any
 - D. With any
-
178. How many bits that we can find in the decimal number B
20 in binary system?
- A. 4
 - B. 5
 - C. 6
 - D. 7
-
179. After a sequential file is updated, the ____ file contains A
the most current data.
- A. New master
 - B. Old master
 - C. Transactio
 - D. Error report
-
180. A ____ defines creates and maintains a database. This C
allows controlled access to data in the database.
- A. Database
 - B. Procedures
 - C. Database management system
 - D. Software

-
181. B



A modern electronic computer is a machine that is meant for

- A. Doing quick mathematical calculations
- B. Input storage, manipulation of outputting of data
- C. Electronic data processing
- D. Performing repetitive tasks accurately
- E. None of the others

182. In storing audio, the quantized sample values need to be ___ as bit patterns

- A. Quantization
- B. Sampled
- C. Decoded
- D. Encoded

183. In the hierarchical database model, data is organized as an inverted ___

- A. Graph
- B. Tree
- C. Array
- D. Record

184. The RSA algorithm for confidentiality uses ___ cryptography. A. Asymmetric-key

- A. Asymmetric-key
- B. Symmetric-key
- C. Substitution
- D. Transposition

185. The complete picture of data stored in database is known as

- A. Record
- B. Schema



- C. System flowchart
- D. DBMS
- E. None of the others

186. The base of the hexadecimal number system is ____ D

- A. 2
- B. 8
- C. 10
- D. 16

187. In a ____ a character (or a symbol) in the plaintext is always changed to the same character (or symbol) in the ciphertext regardless of its position in the text (Chose 2 answers) AB

- A. Monoalphabetic cipher
- B. Substitution cipher
- C. Polyalphabetic cipher
- D. Transposition cipher

188. A computer has 32 MB (megabytes) of memory. How many bits are needed to address any single byte in memory? C

- A. 5 bits
- B. 20 bits
- C. 25 bits
- D. 32 bits

189. Before transmission the sender transforms the plain-text into the ciphertext-this is ____ C

- A. Plain text
- B. Cipher text
- C. Encryption
- D. Decryption



-
190. A notation used to express clearly on algorithm is known as A
- A. Algorithmic language
 - B. Assembly language
 - C. Machine language
 - D. High level language
 - E. None of the others
-
191. ___ is a protocol for file transfer. A
- A. FTP
 - B. SMTP
 - C. TELNET
 - D. HTTP
-
192. What is a program of UNIX that accepts requests from users (processes) and interprets them for the rest of the operating system? C
- A. Software
 - B. Function
 - C. SHELL
 - D. Feature
-
193. Where does a computer add and compare data? C. CPU chip
- A. Hard disk
 - B. Floppy disk
 - C. CPU chip
 - D. Memory chip
 - E. None of the others
-
194. What is the result of storing +25 in an 8-bit memory location using two's complement representation? A
- A. 00011001



- B. 00110010
- C. 10011001
- D. 10110010

195. In the ___ graphic method of representing an image in a computer, each pixel is assign. A

- A. Bitmap
- B. Vector
- C. Quantized
- D. Binary

196. Which of the following is not a type of Artificial Intelligence agent? B

- A. Learning AI agent
- B. Unity-based AI agent
- C. Goal-based AI agent
- D. Simple reflex AI agent

197. ___ is a process in which an algorithm calls itself. C

- A. Insertion
- B. Searching
- C. Recursion
- D. Iteration

198. A ___ is a nonactive set of instructions stored on disk (or tape). It may or may not become a job. A

- A. Program
- B. Job
- C. Process
- D. Task

199. The TCP/IP model has ___ layers. A

- A. Five



- B. Six
 - C. Seven
 - D. Eight
-

200. How many connections (in the data bus) do we need to transfer 4 bytes of a word at the same time? D
- A. 4
 - B. 8
 - C. 16
 - D. 32
-

201. In an inorder traversal of a binary tree, the root is processed B
- A. First
 - B. Second
 - C. Last
 - D. Two times
-

202. ___ is a stand-alone storage location that holds data temporarily. B
- A. An ALU
 - B. A register
 - C. A control unit
 - D. A tape drive
-

203. When a sequential file is updated, the ___ file contains a list of all errors occurring during the update process. D
- A. New master
 - B. Old master
 - C. Transaction
 - D. Error report
-



-
204. ___ tests the functionality of the software in terms of what the software is supposed to accomplish such as its inputs and outputs. (Chose 2 answers) CD
- A. Testing phase
 - B. White-box testing
 - C. Black-box testing
 - D. Boundary testing
-
205. A ___ is a file of characters. It cannot contain integers floating-point numbers or any other .data structures in their internal memory format A
- A. Text file
 - B. Binary file
 - C. Sequential file
 - D. EOF
-
206. Use a simple left shift operation on the bit pattern 1001 1000 C
- A. 0001 1001
 - B. 0001 1010
 - C. 0011 0000
 - D. 0011 0001
-
207. In data storage, text is an example of ___ data and audio is an example of ___ data. B
- A. Analog, digital
 - B. Digital, analog
 - C. Analog, bit pattern
 - D. Digital, quantized
-
208. What is statement that stores the value in the variable, which has already been created in the declaration section. We use the symbol <-- in our algorithm to define B



it?

- A. Compound statement
- B. Assignment statement
- C. Control statement
- D. None of above

209. Conversion of binary number 1010101000010111 to hexadecimal number is ____ C

- A. D8F9
- B. A8B9
- C. AA17
- D. D9F8
- E. None of the others

210. A ____ is the event that occurs when a hashing algorithm produces an address for an insert key but that address is already occupied D

- A. Open addressing
- B. Every possible key
- C. Digit extraction key
- D. Collision

211. ____ ciphers are sometimes called public-key ciphers B

- A. Symmetric-key
- B. Asymmetric-key
- C. Public-key
- D. Private-key

212. The ethical principle that dictates that a decision should be made according to the universally accepted principles of morality is referred to as ____ A

- A. Moral rules
- B. Utilization



- C. Social contract
- D. None of the others

-
213. In ___ paradigm, the service provider is an application program, it runs continuously, waiting for another application program, then to make a connection through the Internet and ask for service
- A. P2P
 - B. Tranditional
 - C. Client-server
 - D. Model
-
214. The TCP/IP suite does not define any protocol in
- A. Transport
 - B. Network
 - C. Data-link
 - D. Session
-
215. Which of the following translate back from machine code something resembling the source language?
- A. Interpreter
 - B. Compiler
 - C. Assembler
 - D. Decompiler
 - E. None of the others
-
216. ___ hides all the details of an algorithm in an attempt to give the 'big picture' and to show how the algorithm flows from beginning to end.
- A. Pseudocode
 - B. Unified Modeling Language (UML)
 - C. Flowchart
 - D. State diagram



-
217. A data bus, the word is 4 bytes, so we need a data bus with ___ connections? D
- A. 4
 - B. 8
 - C. 16
 - D. 32
-
218. The ___ (approximately 1975-1985) saw the appearance of microcomputers D
- A. First generation
 - B. Second generation
 - C. Third generation
 - D. Fourth generation
-
219. A binary search tree (BST) is a binary tree with one extra property: the key value of root is ___ values of all nodes in each left subtree A
- A. Greater than
 - B. Less than
 - C. Equal
 - D. Smaller than
-
220. A/an ___ is a program in a procedural language that is a set of statements. The statements are normally executed one after another. However, sometimes it is necessary to change this sequential order. This figure illustrates for this statement. B
- A. Compound statement
 - B. Control statement
 - C. None of others
 - D. Assignment statement
-



-
221. In the ___ hashing method, there are no synonyms or collisions. A. Direct
- A. Direct
 - B. Division remainder
 - C. Modulo division
 - D. Digit extraction
-
222. The ___ layer of the TCP/IP protocol suite is responsible for source-to-destination delivery of the entire message. A
- A. Transport
 - B. Network
 - C. Data-link
 - D. Session
-
223. In cryptography ___ is ordinary readable text before being encrypted into ciphertext or after being deciphered. A
- A. Plain text
 - B. Cipher text
 - C. Encryption
 - D. Decryption
-
224. The first electronic digital computer contained? B
- A. Electronic valves
 - B. Vacuum tubes
 - C. Transistors
 - D. Semiconductor memory
 - E. None of the others
-
225. Computers are extremely fast and have fantastic memories. However, the only thing they can remember is B



- A. Instruction
- B. Series of 1s and 0s
- C. Boolean algebra
- D. Logic theorems
- E. None of the others

226. A particular system that contains intelligent agents. B. Agency

- A. AI systems
- B. Agency
- C. Autonomous systems
- D. Company

227. RAM is used as a short memory because it is A

- A. Volatile
- B. Has small capacity
- C. Is very expensive
- D. Is programmable
- E. None of the others

228. Given a sound clip of a person or people speaking, determine the textual representation of the speech. B

- A. Text-to-speech
- B. Speech-to-text
- C. Sound-to-text
- D. All of the above
- E. None of the above

229. In a three-level DBMS architecture, the ___ level inter- A
acts directly with the users.

- A. External
- B. Conceptual
- C. Internal
- D. Physical



-
230. In the modulo division hashing assume we have key = A 614, file size = 307. The result of address is ____
- A. 001
 - B. 002
 - C. 003
 - D. 004
-
231. If we call the bit depth or number of bits per sample B, the number of samples per second S, calculate BIT RATE (R)?
- A. $B \times 2S$
 - B. $2B \times S$
 - C. $B \times S$
 - D. $B^2 \times S^2$
-
232. Assume memory location that can only hold four bits. What is the result, if we have stored the integer 11 in a memory location and then try to add 9 to the unsigned integer?
- A. 2
 - B. 9
 - C. 11
 - D. None of the others
-
233. What is taxonomy of computer organization?
- A. SIMD - Single Instruction-stream, Multiple Data-stream
 - B. MISD - Multiple Instruction-stream, Single Data-stream
 - C. MIMD - Multiple Instruction-stream, Multiple Data-stream
 - D. All of others



-
234. If personal information about each employee in a company is stored in a file we can use ___ to retrieve each record at the end of the month to print the pay-checks B
- A. Hashed file
 - B. Sequential file
 - C. EOF
 - D. Index file
-
235. Use a arithmetic left shift operation on the bit pattern 1001 1001 A
- A. 0011 0010
 - B. 1100 1100
 - C. 1101 0010
 - D. 1001 1000
-
236. Which of these terms best describes the type of AI used in today's email spam filters, speech recognition, and other specific applications? A,Artificial Narrow Intelligence (ANI)
- A,Artificial Narrow Intelligence (ANI)
 - B,Artificial General Intelligence (AGI)
-
237. What do you call the commonly used AI technology for learning input (A) to output (B) mappings? B, Supervised learning
- A, Unsupervised learning
 - B, Supervised learning
 - C, Artificial General Intelligence
 - D, Reinforcement learning
-

238.



You want to use supervised learning to build a speech recognition system. The figure above suggests that in order for a neural network (deep learning) to achieve the best performance, you would ideally use: (Select all that apply)

A, A large dataset (of audio files and the corresponding text transcript)
C, A large neural network

A, A large dataset (of audio files and the corresponding text transcript)

B, A small dataset (of audio files and the corresponding text transcript)

C, A large neural network

D, A small neural network

239. The only way to acquire data for a supervised learning algorithm is to manually label it. I.e., given the input A, to ask a human to provide B.

B, False

A, True

B, False

240. Which of these statements regarding data acquisition do you agree with?

A, Some types of data are more valuable than others; working with an AI team can help you figure out what data to acquire.

A, Some types of data are more valuable than others; working with an AI team can help you figure out what data to acquire.

B, It doesn't matter how data is acquired. The more data, the better.

C, Only structured data is valuable; AI cannot process unstructured data.

D, It doesn't help to give data to an AI team, because they can always produce whatever they need by themselves.



-
241. You run a company that manufactures scooters. Which of the following are examples of unstructured data? (Select all that apply.)
- A, Pictures of your scooters
 - B, Audio files of the engine sound of your scooters
 - C, The number of scooters sold per week over the past year
 - D, The maximum speed of each of your scooters
- A, Pictures of your scooters
B, Audio files of the engine sound of your scooters
-
242. Suppose you run a website that sells cat food. Which of these might be a good result from a data science project? (Select all that apply.)
- A, A large dataset of images labeled as "Cat" and "Not Cat"
 - B, Insights into how to market cat food more effectively, depending on the breed of cat.
 - C, A neural network that closely mimics how cats' brains work.
 - D, A slide deck presenting a plan on how to modify pricing in order to improve sales.
- B, Insights into how to market cat food more effectively, depending on the breed of cat.
D, A slide deck presenting a plan on how to modify pricing in order to improve sales.
-
243. Based on the terminology defined in Lesson 4 "The terminology of AI", which of the following statements do you agree with? (Select all that apply.)
- 1 / 1 point
- A, The terms "deep learning" and "neural network" are used almost interchangeably.
 - B, AI is a type of deep learning. (I.e., All AI algorithms are deep learning algorithms.)
 - C, The terms "machine learning" and "data science" are used almost interchangeably.
- A, The terms "deep learning" and "neural network" are used almost interchangeably.
D, Deep learning is a type of machine learning. (I.e., all deep learning algorithms are machine learning algorithms.)
-



D, Deep learning is a type of machine learning. (I.e., all deep learning algorithms are machine learning algorithms.)

-
244. Which of these do AI companies do well? All of the above
- Strategic data acquisition
 - Invest in unified data warehouses
 - Spot automation opportunities
 - All of the above
-
245. Say you want to input a picture of a person's face (A), and output whether or not they are smiling (B). Because this is a task that most humans can do in less than 1 second, supervised learning can probably learn this A-to-B mapping. True
- True
False
-
246. Machine learning is an "iterative" process, meaning that an AI team often has to try many ideas before coming up with something that's good enough, rather than have the first thing they try work. A, True
- A, True
B, False
-
247. Say you want to use Machine Learning to help your sales team with automatic lead sorting. I.e., Input A (a sales prospect) and output B (whether your sales team should prioritize them). The 3 steps of the workflow, in scrambled order, are: C (ii) (iii) (i)
- (i) Deploy a trained model and get data back from



users

(ii) Collect data with both A and B

(iii) Train a machine learning system to input A and output B

What is the correct ordering of these steps?

A (ii) (i) (iii)

B (i) (iii) (ii)

C (ii) (iii) (i)

D (i) (ii) (iii)

248. What are the key steps of a Data Science project?

D All of the above

A Collect data

B Analyze the data

C Suggest hypothesis or actions

D All of the above

249. Machine learning programs can help: (select all that apply)

A Customize product recommendations

B Automate lead sorting in sales

C Automate visual inspection in a manufacturing line

D Automate resume screening

A Customize product recommendations

B Automate lead sorting in sales

C Automate visual inspection in a manufacturing line

D Automate resume screening

250. Unless you have a huge dataset ("Big Data"), it is generally not worth attempting machine learning or data science projects on your problem.

A True

B False

B False



251. **ay you want to build an AI system to help recruiters with automated resume screening. Which of these steps might be involved in the "technical diligence" process? (Select all that apply.)**
- A Making sure you can get enough data for this project
- B Ensuring that this is valuable for your business (e.g., estimating the project ROI)
- C Defining an engineering timeline
- D Making sure that an AI system can meet the desired performance
252. **Which of these statements about "business diligence" do you agree with?**
- 1 / 1 point
- A Business diligence is the process of ensuring that the envisioned AI technology is feasible.
- B Business diligence is the process of ensuring that the AI technology, if it is built, is valuable for your business.
- C Business diligence can typically be completed in less than a day.
- D Business diligence applies only if you are launching new product lines or businesses.
253. **You want to use supervised learning for automated resume screening. Which of the following statements about the training set are true? (Select all that apply.)**
- 1 / 1 point
- A It should give examples of the input A (resume) but not necessarily the desired output B (whether to move
- A Making sure you can get enough data for this project
- C Defining an engineering timeline
- D Making sure that an AI system can meet the desired performance
- B Business diligence is the process of ensuring that the AI technology, if it is built, is valuable for your business.
- B It will be used by the AI team to train the supervised learning algorithm.
- D It should give examples of both the input A (resume) and the desired



forward with a candidate).

output B (whether to move

B It will be used by the AI team to train the supervised learning algorithm. forward with a candidate).

C The training set and test set can be the same dataset.

D It should give examples of both the input A (resume) and the desired output B (whether to move forward with a candidate).

254. For your automated resume screening application, you are now providing a test set to the AI team. Which of the following statements about the test set are true? (Select all that apply.) A It will be used by the AI team to evaluate the performance of the algorithm. B It should give examples of both the input A (resume) and the desired output B (whether to move forward with a candidate) C The test set should ideally be identical to the training set. D It should give examples of the input A (resume) but not necessarily the desired output B (whether to move forward with a candidate).

A It will be used by the AI team to evaluate the performance of the algorithm.

B It should give examples of both the input A (resume) and the desired output B (whether to move forward with a candidate)

255. Which of these are reasons that it's often unrealistic to expect an ML system to be 100% accurate? D All of the above.

A You might not have enough data

B Data can be mislabeled

C Data can be ambiguous

D All of the above.

256.

False



Because a smart speaker can carry out multiple functions (such as tell a joke, play music, etc.) it is an example of Artificial General Intelligence (AGI).

True

False

257. Question 2

What are the key steps to a smart speaker function?

A. Speech recognition 'Trigger word detection -> intent recognition -> command execution.

B. Trigger detection -> intent recognition -> speech recognition -> command execution.

C. Trigger word detection -> speech recognition -> intent recognition -> command execution.

D. Trigger word detection -> intent recognition -> speech recognition -> command execution.

C. Trigger word detection
-> speech recognition ->
intent recognition -> com-
mand execution.

258. Question 3

Consider this system for building a self-driving car:
The component for pedestrian detection is usually built using:

A. A motion planning algorithm

B. GANs

C. Supervised learning

D. Reinforcement learning

C. Supervised learning

259. Suppose you are building a trigger word detection system, and want to hire someone to build a system to map from Inputs A (audio clip) to Outputs B (whether the trigger word was said), using existing AI technology. Out of the list below, which of the following hires would be most suitable for writing this software?

B. Machine learning engineer



- A.AI Product Manager
- B.Machine learning engineer
- C.Machine learning researcher
- D.Data engineer

260. What is the first step in the AI Transformation Playbook for helping your company become good at AI?

B.Execute pilot projects to gain momentum

- A.Provide broad AI training
- B.Execute pilot projects to gain momentum
- C.Build an in-house AI team
- D.Develop an AI strategy

261. According to the AI Transformation Playbook, what is the most important outcome of your first pilot project?

A.Succeed and show traction within 6-12 months

- A.Succeed and show traction within 6-12 months
- B.Drive extremely high value for the business
- C.Be executed by an in-house team
- D.None of the above

262. Say you are building a smart speaker, and want to accumulate data for your product through having many users. Which of these represents the "Virtuous circle of AI" for this product?

(D) <https://imgur.com/9Nasv1N.png>

- (A) supervised learning
- (B) <https://imgur.com/9nasv0n.png>
- (C) trigger word detection -> speech recognition -> intent recognition -> command execution.
- (D) <https://imgur.com/9Nasv1N.png>

263. Why is developing an AI strategy NOT the first step in the AI Transformation Playbook?

C.Without having some practical AI experience and knowing what it feels

- A.The strategy should be to use the Virtuous Circle of



AI, which comes after building a product.

B. When transforming a company into an AI company, one does not need a strategy, therefore it can't be the first step. like to build an AI project, a company usually does not know enough to formulate a sound strategy.

C. Without having some practical AI experience and knowing what it feels like to build an AI project, a company usually does not know enough to formulate a sound strategy.

D. There is no reason. Developing an AI strategy IS the first step in the AI Transformation Playbook.

264. According to the AI Transformation Playbook, broad AI training needs to be provided not only to engineers, but also to executives/senior business leaders and to leaders of divisions working on AI projects. True

True
False

265. Which of the following are AI pitfalls to avoid? (Select all that apply)
- A. Pairing engineering talent with business talent to identify feasible and valuable projects.
- B. Expecting AI to solve everything. B. Expecting AI to solve everything.
- C. Expecting traditional planning processes to apply without changes. C. Expecting traditional planning processes to apply without changes.
- D. Expecting AI based projects to work the first time. D. Expecting AI based projects to work the first time.

266. What are the current limitations of AI technology? (Select all that apply)
- 1 / 1 point
- A. Explainability is hard
- B. AI technology can be biased
- A. Explainability is hard
- B. AI technology can be biased
- C. AI technology is susceptible to adversarial at-



- C.AI technology is susceptible to adversarial attacks
- D.There are no limitations to AI technology
- E.AI technology can display discriminatory behavior

tacks

E.AI technology can display discriminatory behavior

267. What is the Goldilocks Rule of AI?
- A.Fearing human extinction due to AI development
 - B.One shouldn't be too optimistic or too pessimistic about AI technology
 - C.Expecting AI to create Utopia
 - D.AI's technology will continue to grow and will solve all of humanity's problems

B.One shouldn't be too optimistic or too pessimistic about AI technology

268. Say you are building an AI system to help make diagnoses from X-ray scans. Which of the following statements about explainability of AI do you agree with?
- A.Most AI systems are highly explainable, making it easy for a doctor to understand and give a particular diagnosis.
 - B.Lack of explainability can hamper users' willingness to trust and adopt an AI system.
 - C.Explainability is usually achieved through building a chatbot to talk to the user to explain its outputs.
 - D.AI systems are intrinsically "black box" and cannot give any explanation for their outputs.

B.Lack of explainability can hamper users' willingness to trust and adopt an AI system.

269. Using current AI technology, if a machine learning system learns from text containing unhealthy biases/stereotypes, then the resulting AI software tends to exhibit similarly unhealthy biases/stereotypes.

True

True
False



-
270. Using current AI technology, if a machine learning system learns only from text that is completely neutral and does not reflect any gender biases, then we would expect it to exhibit no, or at most minimal, gender bias.
True
False
-
271. What are recommended practices for mitigating bias in AI? (Select all that apply)
- A. Using more inclusive/less biased data
 - B. Systematic auditing processes to check for bias
 - C. Technical solution such as "zeroing out" bias
 - D. Using an adversarial attack on the AI system to change its outputs to be less biased
- A. Using more inclusive/less biased data
B. Systematic auditing processes to check for bias
C. Technical solution such as "zeroing out" bias
-
272. Which of these are examples of adversarial attacks on an AI system? (Select all that apply)
- A. Subtly modifying an audio clip to make a speech recognition system think someone said "Yes, authorized" when they actually said "No, reject."
 - B. Adding a sticker to a stop sign to make an AI system fail to detect it.
 - C. Subtly altering an image to mislead an AI system into mistakenly recognizing a dog as a cat.
 - D. Using AI to synthesize a fake video of a politician saying something they never actually said.
- A. Subtly modifying an audio clip to make a speech recognition system think someone said "Yes, authorized" when they actually said "No, reject."
B. Adding a sticker to a stop sign to make an AI system fail to detect it.
C. Subtly altering an image to mislead an AI system into mistakenly recognizing a dog as a cat.
-
273. If a developing economy has a strong and thriving coffee bean manufacturing industry (or any other ver-
- True



tical industry), then it has an advantage in applying AI to that that industry.

True

False

274. In the decade or so, which jobs are most likely to be displaced by AI? B. Jobs that primarily involve routine, repetitive tasks

A. Most jobs related to office work (white collar jobs)

B. Jobs that primarily involve routine, repetitive tasks

C. Jobs that mainly involve non-routine, non-repetitive tasks

D. All jobs will be displaced

275. A step-by-step solution to a problem is called _____. D. An algorithm

A. Hardware

B. An operating system

C. A computer language

D. An algorithm

276. The first microprocessors produced by Intel Corpn. and Texas Instruments were used primarily to control small C. Calculators

A. Microwave ovens

B. Washing machines

C. Calculators

D. Personal computers

E. Robotics

277. Off-line operation is the operation of devices without the control of _____ B. CPU

A. Memory

B. CPU



- C. ALU
- D. Control unit
- E. None of the others

278. ____ is a memory type with capacitors that need to be refreshed periodically.

B. DRAM

- A. SRAM
- B. DRAM
- C. ROM
- D. CROM

279. In magnetic disks, the ____ defines how fast the disk is spinning.

C. Rotational speed

- A. Seek time
- B. Transfer time
- C. Rotational speed
- D. None of others

280. What is the name given to the molecular-scale computer?

B. Nanocomputer

- A. Femtocomputer
- B. Nanocomputer
- C. Supercomputer
- D. Microcomputer
- E. None of the others

281. The base of the binary number system is ____

A. 2

- A. 2
- B. 8
- C. 10
- D. 16

282.

A. 1011.1011



The binary equivalent of the Octal number 13.54 is

- A. 1011.1011
- B. 1101.1110
- C. 1001.1110
- D. All of the others
- E. None of the others

283. In a positional number system with base b , we can always find the number of digits of an integer. So how many bits can we find in the decimal number 20 in hexadecimal system? B. 2

- A. 1
- B. 2
- C. 3
- D. 4
- E. None of the others

284. In the sign-and-magnitude representation, the memory location can hold four bits, assume that we have stored integer -6 in a memory location and we then try to subtract 7 to the integer. The computer's response is ____? C. 7

- A. 6
- B. -6
- C. 7
- D. -7

285. Different sets of bit patterns have been designed to represent text symbols. Each set is called a ____? C. code

- A. bit patterns
- B. binary



- C. code
- D. binary text file

-
286. When we want to store music in a computer, the audio signal must be _____.
A. Sampled only
B. Quantized only only
C. Coded only
D. Sampled, quantized, and coded
D. Sampled, quantized, and coded
-
287. What is NOT a standard of vector graphic?
A. CAD
B. FLASH
C. TrueType
D. PostScript
E. None of others
E. None of others
-
288. Use a simple left shift operation on the bit patter 1001 1000
A. 0001 1001
B. 0001 1010
C. 0011 0000
D. 0011 0001
C. 0011 0000
-
289. A ____ WAN is a network that connects two communicating devices through a transmission medium (cable or air)
A. Peer-to-peer
B. Node-to-node
C. Point-to-point
D. Data link
C. Point-to-point
-
290. B. Port numbers



At the transport layer, addresses are called _____ and these define the application-layer programs at the source and destination.

- A. TCP
- B. Port numbers
- C. UDP
- D. IP address

291. _____ is the standard protocol provided by TCP/IP for copying a file from one host to another and transferring files from one system to another seems simple and straightforward.

A. FTP

- A. FTP
- B. SMTP
- C. URL
- D. HTTP

292. All the instructions and procedures needed to start up a computer, to the point it can load an operating system, are stored in ROM chips. In IBM compatible PCs this is called the _____

B. ROM BIOS

- A. RAM BIOS
- B. ROM BIOS
- C. EPROM
- D. POST
- E. None of the others

293. Which is an electronic device that can store temporarily a single bit of data

D. Latch

- A. Accumulator
- B. Buffer



- C. Memory
- D. Latch

294. Which technique is used in multiprogramming that needs to be determined size beforehand by the memory manager and with each program occupying a contiguous space? A. Partitioning

- A. Partitioning
- B. Paging
- C. Demand paging
- D. Demand segmentation

295. A ____ is a program in execution. It is a program that has started but has not finished C. Process

- A. Program
- B. Job
- C. Process
- D. Task

296. The list of coded instructions is called A. Computer program

- A. Computer program
- B. Algorithm
- C. Flowchart
- D. Utility programs
- E. None of the others

297. The ____ analyzer checks the sentences created by the syntax analyzer to be sure that they contain no ambiguity. C. Semantic

- A. Lexical
- B. Syntax
- C. Semantic
- D. Code generation



-
298. An instruction that transfers program control to one or more possible paths is known as
- C. Broadband channel
- A. Utility program
 - B. System software
 - C. Broadband channel
 - D. Application program
 - E. None of the others
-
299. What is language of procedural paradigm that is high level language and also has some low-level instructions?
- B. C
- A. COBOL
 - B. C
 - C. Pascal
 - D. Ada
-
300. A _____ causes an action to be performed by the program. It translates directly into one or more executable computer instructions.
- D. Statements
- A. Expression
 - B. Operator
 - C. Variables
 - D. Statements
-
301. What is the analysis process in the system implementation phase that will use a program as an active agent that manipulates passive object?
- D. All of others
- A. Procedure-oriented analysis
 - B. Structured analysis
 - C. Classical analysis
 - D. All of others
-



-
302. One phase in system development is _____. A. Analysis
B. Application
C. Designing
D. Collecting
-
303. _____ is the breaking up of a large project into smaller parts. D. Modularization
A. Coupling
B. Incrementing
C. Obsolescence
D. Modularization
-
304. In the software quality, what factor measures can be mentioned for transferability? C. Resusability
A. Accuracy
B. Timeliness
C. Resusability
D. Changeability
-
305. What is method in black-box testing which tests are designed to include representatives of boundary values in a range? A. Boundary-value testing
A. Boundary-value testing
B. Random testing
C. Exhaustive testing
D. Data flow testing
-
306. An array that consists of just rows and columns is a ____ array. B. Two-dimensional
A. One-dimensional
B. Two-dimensional
-



- C. Three-dimensional
- D. Multidimensional

-
307. The name of a linked list is the name of the head pointer that points to the ____ of the list. A. First node
- A. First node
 - B. Second node
 - C. Null node
 - D. Last node
-
308. The pop operation ____ of the stack. A. Deletes an item from the top
- A. Deletes an item from the top
 - B. Deletes an item from the bottom
 - C. Inserts an item at the top
 - D. Inserts an item at the bottom
-
309. In preorder traversal of a binary tree, the ____ C. Root is processed first
- A. Left subtree is processed first
 - B. Right subtree is processed first
 - C. Root is processed first
 - D. The root is never processed
-
310. The enqueue operation ____ an item at the ____ of the queue. D. Insert/rear
- A. Delete/front
 - B. Delete /end
 - C. Inserts/front
 - D. Insert/rear
-
311. The address produced by a hashing algorithm is the ____ address. D. Home
- A. Probe
 - B. Synonym



- C. Collision
 - D. Home
-

312. In the relational database model, data is organized in ____ called relations. B. Two-dimensional tables
- A. Two-dimensional array
 - B. Two-dimensional tables
 - C. Two-dimensional record
 - D. None of others
-
313. If you want to change the value of an attribute of a tuple, you use the ____ operation. C. Update
- A. Project
 - B. Join
 - C. Update
 - D. Select
-
314. ____ is the process by which a given set of relations are transformed to a new set of relations with a more solid structure. D. Normalization
- A. Relational database
 - B. Database system
 - C. Atomic operations
 - D. Normalization
-
315. Which of the following attacks is threatening confidentiality? A. Snooping
- A. Snooping
 - B. Repudiation
 - C. Denial of service
 - D. Modification
-

316.

B. Steganography



_____ means concealing the message by covering it with something else.

- A. Cryptography
- B. Steganography
- C. Compressing
- D. Authentication

317. _____ is an attack that reduce the capability of a computer system to function correctly or bring the system down altogether by exhausting its resources.

A. DoS

- A. DoS
- B. Trojan horses
- C. Worms
- D. Viruses

318. In a block cipher a _____ is used to encrypt the whole block even if the key is made of multiple values

B. Single key

- A. Key block
- B. Single key
- C. Key stream
- D. Auto key

319. Use a arithmetic right shift operation on the bit patter 1001 1000

B. 1100 1100

- A. 0100 1100
- B. 1100 1100
- C. 1101 1001
- D. 1001 1000

320. A 17th-century computing machine that could perform addition and subtraction was the

A. Pascaline

- A. Pascaline
- B. Jacquard loom



- C. Analytical Engine
 - D. Babbage machine
-

321. After the instruction is decoded, the control unit sends the task order to a component in the CPU. What is this phase? C. Excute phase

- A. Fetch phase
 - B. Decode phase
 - C. Excute phase
 - D. None of others
-

322. For surface organization of magnetic tape, the width of the tape is divided into nine tracks. Nine vertical locations can store 8 bits of information related to a byte plus a bit for D. Error detection

- A. Next block
 - B. Data
 - C. Next tape information
 - D. Error detection
-

323. IBM-PC is an example of C. Micro computer

- A. A main frame
 - B. Machine-in-built
 - C. Micro computer
 - D. Special purpose
 - E. None of the others
-

324. Instructions and memory addresses are represented by B. Binary codes

- A. Character codes
- B. Binary codes
- C. Binary word



- D. Parity bit
- E. None of the others

325. What is the result of storing -25 in an 8-bit memory location using two's complement representation? C. 11100111|

- A. 00011001
- B. 00110010
- C. 11100111|
- D. 11110010

326. The ASCII code for the character J is: C. 0100 1010

- A. 1001 0001
- B. 1001 1010
- C. 0100 1010
- D. 1010 0001
- E. None of the others

327. If we call the bit depth or number of bits per sample 8, the number of samples per second S, so how to calculate BIT RATE (R)? C. $B \times S$

- A. $B \times 2S$
- B. $28 \times S$
- C. $B \times S$
- D. $B^2 \times S^2$

328. Use a arithmetic right shift operation on the bit pattern 1001 1000 B. 1100 1100

- A. 0100 1100
- B. 1100 1100
- C. 1101 1001
- D. 1001 1000

329. C. 0011 0000



Use a simple left shift operation on the bit pattern

1001 1000

- A. 0001 1001
- B. 0001 1010
- C. 0011 0000
- D. 0011 0001

330. The format of an Ipv6 address is referred to as _ D. 8- Hexadecimal digits

- A. Dotted-hexadecimal notation
- B. Dotted-octal notation
- C. Dotted-decimal notation
- D. 8- Hexadecimal digits

331. explicitly defines connection establishment, data transfer, connection teardown phases to provide a connection-oriented service AND reliable protocol A. TCP

- A. TCP
- B. UDP
- C. IP
- D. HTTP

332. In a_ the directory system listing of the peers and what they offer uses the client-server paradigm, but the storing and downloading of the files are done using the peer-to-peer paradigm. B. Centralized network

- A. WWW
- B. Centralized network
- C. Hybrid network
- D. Decentralized network

333. Which of the following is the user programmed semi-conductor memory? C. EPROM

- A. SRAM



- B. DRAM
- C. EPROM
- D. All of the others
- E. None of the others

334. Which of the following holds data and processing instructions temporarily until the CPU needs it? C. Main memory

- A. ROM
- B. Control unit
- C. Main memory
- D. Coprocessor chips
- E. None of the others

335. In____sort, the smallest item from the unsorted list is swapped with the item at the beginning of the unsorted list. A. Selection

- A. Selection
- B. Bubble
- C. Insertion
- D. Every

336. The program in a high-level language is called the B. Source program

- A. Programming language
- B. Source program
- C. Translation
- D. Interpretation

337. The code that should be adequate for present and anticipated data processing both for machine and human use is C. Operability

- A. Conciseness
- B. Uniqueness



- C. Operability
- D. All of the others

-
338. _ are names for memory locations A. Variables
- A. Variables
 - B. Imperative language
 - C. Identifiers
 - D. Data type
-
339. between modules in a software system must be maximized. B. Cohesion
- A. Coupling
 - B. Cohesion
 - C. Neither coupling nor cohesion
 - D. Both coupling and cohesion
-
340. Testing a software system can involve____testing D. Both black-box and glass-box
- A. Black-box
 - B. Glass-box
 - C. Neither black-box nor glass-box
 - D. Both black-box and glass-box
-
341. A data structure can be D. An array, a record, or a linked list
- A. Only an array
 - B. Only a record
 - C. Only a linked list
 - D. An array, a record, or a linked list
-
342. Given a linked list called children, the pointer variable children identifies____element of the linked list. A. The first
- A. The first
 - B. The second



- C. The last
 - D. Any
-

343. When a hashing algorithm produces an address for an insertion key and that address is already occupied, it is called a_____ A. Collision
- A. Collision
 - B. Probe
 - C. Synonym
 - D. Linked list
-

344. The intersection of a column and row in a spreadsheet is called a bon or C. Cell
- A. Key
 - B. Field
 - C. Cell
 - D. Menu
 - E. None of the others
-

345. Which of the following attacks is threatening integrity? A. Masquerading
- A. Masquerading
 - B. Traffic Analysis
 - C. Denial of service
 - D. Encoding
-

346. _ who create algorithms to break existing internet networks so as to solve the loopholes in them C. White Hat Hacker
- A. Black Hat Hacker
 - B. Grey Hat Hacker
 - C. White Hat Hacker
 - D. Blue Hat Hacker
-



347. In a stream cipher encryption and decryption are done one symbol. We have a plaintext stream a cipher-text stream and a
- A. Key block
 - B. Single key
 - C. Key stream
 - D. Auto key
-